

“Does Economic Concept Selection Matter for Monetary Policy Shock Identification?”

Presentation by Huang Zheng — Evidence from Aruoba & Drechsel (2024)

Overview. This presentation replicates Aruoba & Drechsel (2024), which scores 296 economic concepts from Fed Greenbook documents to explain FFR changes — the leftover is the monetary policy shock. Beyond replication, the author asks whether the specific concepts chosen matter, testing this by swapping the human list for an LLM-generated one and running a label-shuffling placebo. The replication is clean; the robustness analysis raises interesting findings but needs clearer framing and stronger follow-through.

1. Replication

The numbers match. All four predictability horizons are identical to the original and first-stage R^2 values across seven specifications are within ± 0.01 . Small differences come from Python vs. R implementations of Ridge, which the author explains clearly. No issues.

2. Results by Fed Chair

Splitting the sample by Fed chair is the most interesting addition to the presentation:

Chair	Period	Meetings	Extra R^2 from Sentiment
Volcker	1982–87	39	+0.16
Greenspan	1987–06	143	+0.05
Bernanke	2006–08	22	+0.02

The big Volcker gain makes sense — discretionary policy means forecasts alone miss a lot, and Greenbook text fills the gap. Under Greenspan and Bernanke, more systematic policy means forecasts already do most of the work. The Bernanke window is only 22 meetings though, so that +0.02 needs a confidence interval before it means much.

3. Human vs. LLM Concept Lists

The LLM selected 595 concepts from the same 2,000-candidate pool, with only 249 matching the human list — 37% disagreement. Humans picked specific terms like *drilling activity*; the LLM also admitted broad ones like *markets* or *growth*. Neither is wrong, they just applied different criteria. The main takeaway is that concept selection is less objective than it looks, which motivates the shuffling test below.

4. Label Shuffling: Results and Interpretation

For each meeting, the author scrambled which concept gets which sentiment score and re-ran the regression. The fit barely moves: the mid-size model goes from $R^2 = 0.64$ to 0.65, the full model drops from 0.96 to 0.93. The reason is straightforward: within any one meeting, all 296 scores already point in the same direction (86% of concepts are pro-cyclical), so shuffling just reassigns near-identical numbers to different names. The document’s overall tone is preserved regardless of labels.

The decomposition below shows what is actually doing the work:

Specification	R^2	ΔR^2
Greenbook forecasts only	0.49	—
+ Document tone (PC1) with lags & squares	0.51	+0.014
+ All 296 individual sentiment series	0.96	+0.455

PC1 — the overall document tone — adds almost nothing even with full lag structure. The 296 individual series together add +0.455. But labels do not matter. The conclusion is that what counts is having enough separate measurement points across the document, not which specific words anchor them.

Recommendations

1. **Show individual shuffle distributions, not just the mean.** Reporting only the average R^2 across permutations could mask variation. The author should plot the full distribution of shock correlations across independent shuffles to confirm the result is not an averaging artefact.
2. **Run the nested PC1 decomposition.** To isolate what the 296 concepts actually contribute beyond overall tone, give PC1 the same lags and nonlinear terms as the full model and compare it against the full 296-series spec. This directly answers whether it is the tone or the concept-level heterogeneity that drives the fit.
3. **Apply the shuffle test to the LLM list.** The LLM list is more balanced (52% pro-cyclical, 42% counter-cyclical), so within-row variation is larger and the test is actually informative. If labels matter more on the LLM list, it confirms that the original test only passed because of the one-sided composition of the human list.
4. **Validate shuffled shocks via impulse responses.** Plug the shuffled-label shocks into a Local Projections framework and compare IRFs across macro variables (unemployment, output, prices, financial conditions) against the original AD2024 shocks. This is the right downstream check — if the economics do not change, the robustness claim is much stronger.
5. **Include an overfitting diagnostic.** With 3,224 variables and 204 observations, readers will worry. A random-drop test is particularly telling: if dropping half the sentiment variables increases R^2 , it shows the variables are redundant rather than informative — which actually supports the sampling-points interpretation. Bootstrapped coefficient stability or noise injection would also help.
6. **Reframe the conclusion.** The presentation currently concludes that “concept selection does not matter.” The evidence is more precise than that: *labels* do not matter, but the *number* of concepts does, because coverage across the document is what the model exploits. This is a more interesting and defensible claim and the decomposition table already supports it.

Overall, this is a strong presentation. The replication is solid, the chair-level breakdown adds real insight, and the shuffling experiment is a genuinely interesting contribution. The main ask is to follow through on what the evidence already implies: the right conclusion is not that concept selection is irrelevant, but that what matters is coverage, not labels. The six points above could possibly be steps to make that case properly.